

**AMENDMENT NO. 3 JULY 2014**  
**TO**  
**IS 12776 : 2002 GALVANIZED STRAND FOR**  
**EARTHING — SPECIFICATION**

*(First Revision)*

(Page 1, clause 4.1) — Substitute the following for the existing clause:

‘4.1 The steel wire shall be drawn from carbon steel wire rods produced by either the acid or the basic open hearth process, the electric furnace process or the basic oxygen process and shall have the following chemical composition:

| <i>Element</i> | <i>Percentage Composition</i> |
|----------------|-------------------------------|
| Carbon         | 0.55, <i>Max</i>              |
| Manganese      | 0.40 to 0.9                   |
| Phosphorous    | 0.04, <i>Max</i>              |
| Sulphur        | 0.04, <i>Max</i>              |
| Silicon        | 0.15 to 0.35’                 |

(Page 3, Annex A, second row) — Substitute the following for the existing:

|                  |              |      |      |      |      |      |      |
|------------------|--------------|------|------|------|------|------|------|
| 2. Wire Size (D) | a) Nominal   | 3.00 | 3.15 | 3.25 | 3.33 | 3.66 | 4.00 |
| (mm)             | b) Tolerance | ± 2% | ± 2% | ± 2% | ± 2% | ± 2% | ± 2% |

(Page 3, Annex A, seventh row) — Substitute the following for the existing:

|                                                             |                                               |                                               |                                               |                                               |                                               |                                               |
|-------------------------------------------------------------|-----------------------------------------------|-----------------------------------------------|-----------------------------------------------|-----------------------------------------------|-----------------------------------------------|-----------------------------------------------|
| 7. Zn-coating on wires                                      |                                               |                                               |                                               |                                               |                                               |                                               |
| a) Amount of coating (g/m <sup>2</sup> ) (before stranding) | <i>Min: 240</i>                               | <i>Min: 240</i>                               | <i>Min: 250</i>                               | <i>Min: 250</i>                               | <i>Min: 260</i>                               | <i>Min: 260</i>                               |
| (after stranding)                                           | <i>Min: 228</i>                               | <i>Min: 228</i>                               | <i>Min: 238</i>                               | <i>Min: 238</i>                               | <i>Min: 247</i>                               | <i>Min: 247</i>                               |
| b) Uniformity of coating (before stranding)                 | <i>Min: 3 dips of 1 min</i>                   | <i>Min: 3 dips of 1 min</i>                   | <i>Min: 3 dips of 1 min</i>                   | <i>Min: 3 dips of 1 min</i>                   | <i>Min: 3 dips of 1 min</i>                   | <i>Min: 3 dips of 1 min</i>                   |
| (after stranding)                                           | <i>Min: 2 dips of 1 min and 1 dip of 30 s</i> | <i>Min: 2 dips of 1 min and 1 dip of 30 s</i> | <i>Min: 2 dips of 1 min and 1 dip of 30 s</i> | <i>Min: 2 dips of 1 min and 1 dip of 30 s</i> | <i>Min: 2 dips of 1 min and 1 dip of 30 s</i> | <i>Min: 2 dips of 1 min and 1 dip of 30 s</i> |

(Page 4, Annex A, eighth row) — Substitute the following for the existing:

|                                                             |                                                             |                                                             |                                                             |                                                             |                                                             |                                                             |
|-------------------------------------------------------------|-------------------------------------------------------------|-------------------------------------------------------------|-------------------------------------------------------------|-------------------------------------------------------------|-------------------------------------------------------------|-------------------------------------------------------------|
| 8. Lay and lay length of galvanized steel earth strand (mm) | RH 12 to 18 times the nominal strand diameter of earth wire | RH 12 to 18 times the nominal strand diameter of earth wire | RH 12 to 18 times the nominal strand diameter of earth wire | RH 12 to 18 times the nominal strand diameter of earth wire | RH 12 to 18 times the nominal strand diameter of earth wire | RH 12 to 18 times the nominal strand diameter of earth wire |
|-------------------------------------------------------------|-------------------------------------------------------------|-------------------------------------------------------------|-------------------------------------------------------------|-------------------------------------------------------------|-------------------------------------------------------------|-------------------------------------------------------------|